Reg. No										

St. Joseph's College of Arts & Science (Autonomous)
Cuddalore – 607001

AMTCM401 – BUSINESS MATHEMATICS

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. Define Partition of A.
2. Find the slope of the line joining P(-2,3) and Q(8,-5).
3. Define a function $f(x)$ continuous at x=a.
4. Define EOQ
5. A state levies 12% sales tax. Find the tax on a car listed at Rs.42,300.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Find $A \times B \times C$ if A={1,2}, B={0,5} and C={6,7}.
7. If A(-3,3), B(5,9) and C(-7,4) find the distance between A and B; B and C.
8. Evaluate $\lim_{n\to\infty} \frac{2n^2+3n+5}{-5n^2+7n+9}$.
9. The relationship between profit P and advertising cost x is given by

   $P = \frac{4000x}{500+x} - x$. Find x which maximizes P
10. Find the sum up to 'n' terms of the series 6+66+666+6666+…

SECTION - C (5X10=50)

Answer **ALL** Questions

11. (a). Show that the De Morgan's Laws.

[OR]

(b). In a survey concerning the smoking habits of consumers it was found that, 55% smoke cigarette A, 50% smoke B, 42% smoke C, 28% smoke A and B, 20% smoke A and C, 12% smoke B and C and 10% smoke all the three cigarettes.

(i) What percentage do not smoke?

(ii) What percentage smoke exactly two bands of cigarettes?

12. (a). Find the ratio in which the join (-5,1) and (1,-3) divides the straight line passing through (3,4) and (7,8).

[OR]

(b). A company estimates that when its sales is Rs.60,000 its variable expense will be Rs. 30,000 for a fixed expense of Rs. 10,000. Find the break-even point. What is the profit when the sales is Rs. 50,000?

13. (a). The total cost, C, of making x units of a product is $C=0.00003x^3-0.045x^2+8x+25{,}000$. Find the marginal cost at 1000 units out-put.

[OR]

(b) Find the marginal revenue (MR) at price 6 units when the demand function is given by $x = p^{0.2}\ e^{-0.3p}$. Prove that MR is 0 when p=0 or p=4.

14. (a) The cost of a truck engine overhaul is Rs.1000 and the cumulative running cost for m miles is $0.03m + 0.00001m^{2.}$.Find

the mileage at which the truck should be overhauled so as to minimize the average cost per mile.

[OR]

(b) A firm has revenue function R, given by R=10q and the cost function C given by $c=10000+50(q/1000)^2$.Find q which maximizes the profit.

15. (a). A discount of 10% on cash sales and 3% on credit sales is given by a dealer who had listed his articles 25% higher than the actual cost. If 3/5 of his stock were sold on cash sales find the percentage of his gross profit.

[OR]

(b). A person borrowed a sum of rupees 3,500 on the condition to repay in 23 monthly installments of Rs. 150 and the last installment of Rs. 50. Calculate the simple interest at the rate of 5% per annum, which he has to pay as the 25th installment.

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